(Please write your Enrollment Number) EnrollmentNo. 02701182025

MID-TERM EXAMINATION
(Course Name: B. Tech. 1st Year) (Semester: 1st)
(October, 2025) OFF LINE mode

Subject Code: BEC 105	Subject: Fundamentals of Electrical Sciences
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**
	a) Define Kirchhoff's Current Law.	
	b) State and Explain Superposition Theorem.	
	c) Derive Expression of Average Value for sinusoidal wave.	
	d) Define Time Constant in RL and RC circuits.	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) State the Thevenin's Theorem. Using this Theorem calculate the current I through the 10 Ω resistance, circuit shown in fig-1.	
	b) State and Prove the Maximum Power Transfer Theorem.	
	c) For the circuit shown in fig-1, determine the current (I) through the 10 Ω resistance using Mesh Analysis Method.	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	a) Derive the impedance expression and Q-factor for a series R-L-C circuit.	
	b) Compare R-L-C series and R- L-C parallel resonance circuits.	
	c) Consider the RLC series circuit shown in Fig. 2 (Vs = 2V), (R = 6 Ω), (L = 2 H), (C = 0.25 F). Determine i(t).	

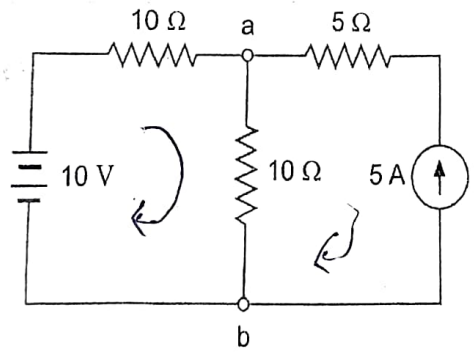


Fig. 1

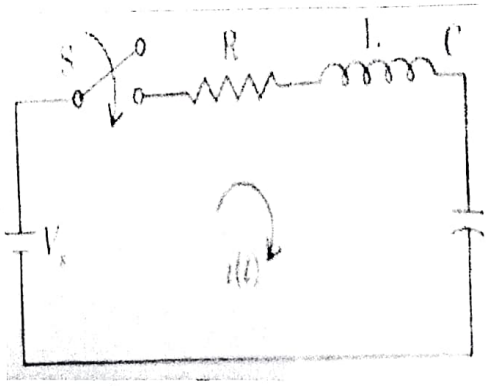


Fig. 2

$10 + 10i + 10(i_1 - i_2) = 0$

$10 + 10i_1 + 10i_1 - 10i_2 = 0$

$10 + 20i_1 - 10(5) = 0$

$20i_1 - 50 = 0$

$20i_1 = 50$

$i_1 = 2$